



FEDERAL BUREAU OF INVESTIGATION

RISI FS-43 Firing System

Requisition Number: SHOP-PR-25-002173

Division(s): Laboratory Division



Limited Source Justification

Nature and/or description of the action being approved

The Federal Bureau of Investigation's LABORATORY DIVISION requires the following product(s)/service(s):

- Industry Specific Machinery

In support of TIEDS mission to safely, reproducibly prepare, and characterize improvised explosive charges and other materials to support law enforcement mission operations, investigations, and core functions.

Contract Type

New contract/agreement; Firm Fixed Price

Statutory authority permitting restricted (other than full and open) competition

The Federal Bureau of Investigation's selected statutory authority which permits restricted (other than full and open) competition for this requirement is listed below.

41 U.S.C. 152(3)/ 40 U.S.C. 501, FAR 8.405-6(a)(1)(i)(B) — Only One Capable Source

Unique qualifications of the proposed contractor or the nature of the action requiring the use of the authority

The Federal Bureau of Investigation's qualification of the selected statutory authority which permits restricted (other than full and open) competition for this requirement is stated below.

41 U.S.C. 152(3)/ 40 U.S.C. 501, FAR 8.405-6(a)(1)(i)(B) — Only One Capable Source

Unique qualification(s) details:

TIEDS works on their dedicated explosive range with Improvised Explosives (IE) and commercial explosives, that are inherently hazardous materials, to conduct explosive characterization research. Since TIEDS inception, in 2015, the five (5) firing systems on the explosive range and detonation chamber are Teledyne RISI. The current Teledyne RISI firing systems were purchased on source documents: (FS-10) 2QA-DJF-15-0700-P-0006762 and (FS-43) 2QA-DJF-16-1200-P-0002597 for an investment of over

To mitigate safety concerns TIEDS uses Exploding Bridge Wire (EBW) blasting caps. Most commercial blasting caps use primary explosives, with a hot wire, which are more sensitive to be set off with less than 9 volts, and providing limited accuracy during explosive characterizing research.

The EBW caps use a gold foil wire and single secondary explosive, requiring higher voltage and being less sensitive. The EBW caps take around 4,000 volts to set off making them less sensitive to static energy, drastically reducing inadvertent discharge, in turn this makes the EBW caps safer to handle on the explosive range. The EBW caps are also recognized and understood in the

explosives research community for several decades. These EBW caps use an FS-43 firing device to initiate the firing detonation. The firing system consists of two parts, FS-43 Control Unit and FS-43 Firing Module, that only work with products from the same manufacture. TIEDS has a complete infrastructure and research methodology designed with 6-pin firing lines, used exclusively with the exact model of FS-43 Control Unit and FS-43 Firing Module, that are ran underground for over 1,000ft between the observation bunk and three (3) instrumentation bunkers.

If any other firing device where to be installed, the entire range complex would have to be rewired due to different manufactures firing line design. Resulting in TIEDS significant downtime of the TIEDS explosive range, cost ten of thousands of dollars for new material to rewire preventing the capability to rapidly respond to National Security threats.

This requirement will expand the current system's capability and replacement of components have met end-of-life.

Anticipated dollar value of the proposed acquisition and determination by the contracting officer that the anticipated cost to the government will be fair and reasonable

The Federal Bureau of Investigation's methodology which supports the determination of fair and reasonable cost to the Government is provided below.

Market Research (including an Independent Government Cost Estimate)

Received an in-formal product quote.

Past Performance Review

TIEDS has been using Teledyne Risi since 2015 and calibration and testing methodologies are specific to this vendor and model.

Description of the market research conducted and the results

The Federal Bureau of Investigation's description of the market research technique(s) it employed to determine the necessity of restricted (other than full and open) competition is provided below.

Reviewing catalogs and other generally available product literature published by manufacturers, distributors, and dealers or available on-line.

Conducted research through catalogs of available products in the market.

Other facts supporting the use of restricted (other than full and open) competition

The Federal Bureau of Investigation's description of additional facts which support the use of restricted (other than full and open) competition is provided below.

FS-43 Firing System (Control Module and Firing Module) requirements to be long distance and remote explosive firing system with input energy of 110-220VAC and output energy 4,000 volt pulse with 1500 amperes peak current to a low resistance load (8.0 joules).

- Shall have a sealed arm switch with indicator light
- adjustable for long control cables
- three-pin plug for 110VAC
- Shall have a meter to indicate module arm voltage and discharge rate
- BNC connector for external trigger IN and OUT
- Short circuit protected
- safety interlock connector
- Six-pin connector for connection from Control Unit
- Five-way binding post output terminals

Statement of any actions the agency may take to remove or overcome any barriers to competition if subsequent acquisitions are anticipated

The Federal Bureau of Investigation's planned and/or executed actions to remove or overcome competitive barriers for subsequent acquisitions are described below.

A competitive procurement is planned in the future.

Any vendor that can supply the specific brand and model requested: Teledyne RISI FS-43.

Description of the products and/or services required to meet the agency's needs

The Federal Bureau of Investigation's description of the requirement which meets its needs is listed below.

Products

Base Period: 2025-04-14 to 2026-04-13

	Quantity	Unit Price	Line Total
Lab Supplies	3 Each		
FS-43 Control Unit			
	Quantity	Unit Price	Line Total
Lab Supplies	5 Each		
FS-43 Module			
	Quantity	Unit Price	Line Total
Lab Supplies	1 Each		
Cable Assembly, 1000ft			

	Quantity	Unit Price	Line Total
Lab Supplies	1 Each		
Processing and Handling			

	Quantity	Unit Price	Line Total
Lab Supplies	1 Each		
Shipping			

Services

Base Period: 2025-04-14 to 2026-04-13

Other Direct Costs (ODCs)

This contract will allow for the purchase of services or goods classified as other direct costs (ODC) as identified herein. Any ODCs not documented in advance of contract award will require written approval by the FBI contracting officer (CO) prior to the purchase. Requests for approval must specifically identify the services or goods to be procured and the estimated price of each. Justification is required for all ODCs that demonstrate that the goods or services to be purchased are in direct support of the contract. Invoices submitted for payment that include ODCs must specifically identify the services or goods purchased and break out in detail the costs associated with the ODCs. Invoices containing ODCs not listed in the contract or identified and approved by the CO in advance of billing will be treated as improper invoices and will not be paid. Similarly, invoices that include ODCs listed in the contract but fail to specifically identify them and elucidate the costs in detail, will be treated as improper and will not be paid.

Base Period: 2025-04-14 to 2026-04-13

